

Sample Paper 3 — Answer Key

Total: 60 marks (45×1 mark + 5×3 marks), 60 minutes.

Section 1 — Logical Reasoning

1. **D) Shirt** — Hammer, Saw, and Nail are tools; a Shirt is not.
2. **A) Ship** — A pilot commands an airplane; a captain commands a ship.
3. **A) VSWI** — The rule is "+4 letters": $R \rightarrow V$, $O \rightarrow S$, $S \rightarrow W$, $E \rightarrow I$.
4. **C) 112** — Each term doubles: 7, 14, 28, 56, 112.
5. **B) 19th** — Position from the left = $32 - 14 + 1 = 19$.
6. **C) West** — Two 90° anticlockwise turns from East = 180° turn = West.
7. **A) 4, 2, 3, 1** — Natural education order: Kindergarten $\rightarrow$ Primary School $\rightarrow$ High School $\rightarrow$ University.
8. **C) 23** — 12, 18, 30 are multiples of 6; 23 is not.
9. **B) Monday** — 31 days = 4 full weeks (28 days, same day: Friday) + 3 more days $\rightarrow$ Sat, Sun, Mon.
10. **B) Cousin** — Sonia's father's brother is her paternal uncle; his son is her cousin.
11. **C) 159** — Each term is "double and add 1": $4 \times 2 + 1 = 9$, $9 \times 2 + 1 = 19$, $19 \times 2 + 1 = 39$, $39 \times 2 + 1 = 79$, $79 \times 2 + 1 = 159$.
12. **A) OQPGA** — The rule is "+2 letters" (Y wraps to A): $M \rightarrow O$, $O \rightarrow Q$, $N \rightarrow P$, $E \rightarrow G$, $Y \rightarrow A$.
13. **C) Square** — Cube, Cuboid, and Pyramid are 3-D solids; a Square is a flat (2-D) shape.
14. **D) Arjun** — Putting it together: Neha > Kabir > Meera > Arjun, so Arjun is lowest.
15. **C) 125** — These are cube numbers: 1^3 , 2^3 , 3^3 , 4^3 , $5^3 = 1, 8, 27, 64, 125$.

Section 2 — Mathematical Reasoning

16. **A) 96,801** — $67,845 + 28,956 = 96,801$.
17. **A) 31,543** — $70,002 - 38,459 = 31,543$.
18. **A) 47,523** — $6,789 \times 7 = 47,523$.
19. **A) 12,558** — $75,348 \div 6 = 12,558$.
20. **B) 9** — HCF of 18 (2×3^2) and 27 (3^3) is $3^2 = 9$.
21. **B) 40** — Multiples of 8: 8, 16, 24, 32, 40... Multiples of 10: 10, 20, 30, 40... First common one = 40.
22. **A) 10/16** — Multiply top and bottom of $5/8$ by 2.
23. **A) 5/6** — Same denominator: $5 + 3 + 2 = 10$, so $10/12$, which simplifies to $5/6$.

24. **A) $\frac{7}{25}$** — $0.28 = \frac{28}{100}$, which simplifies ($\div 4$) to $\frac{7}{25}$.
25. **C) 117** — $65\% \text{ of } 180 = 0.65 \times 180 = 117$.
26. **B) 45%** — $\frac{9}{20} = 0.45 = 45\%$.
27. **C) 25** — 1 part = $40 \div 8 = 5$; smaller = 5 parts = 25.
28. **B) 10%** — Loss = $540 - 486 = ₹54$; loss % = $54/540 \times 100 = 10\%$.
29. **C) ₹1,080** — SI = $(6000 \times 9 \times 2) \div 100 = ₹1,080$.
30. **A) 3,120** — 3 L = 3,000 mL; + 120 mL = 3,120 mL.
31. **B) 3 h 15 min** — $195 \div 60 = 3$ remainder 15.
32. **B) 80 cm** — Perimeter = $2 \times (27 + 13) = 80$ cm.
33. **B) 121 sq cm** — Area = $11 \times 11 = 121$ sq cm.
34. **C) 6** — A regular hexagon has 6 lines of symmetry.
35. **C) 6** — One and a half turns = 540° ; $540 \div 90 = 6$.

Section 3 — Everyday Mathematics

36. **B) 33,300** — $1,850 \times 18 = 33,300$.
37. **A) ₹4,150** — $9500 - 3200 - 2150 = 4,150$.
38. **C) 180 L** — $\frac{3}{4}$ of 720 = 540 L used; $720 - 540 = 180$ L left.
39. **C) ₹30** — $720 \div 24 = 30$.
40. **B) 297** — $11 \times 27 = 297$.
41. **C) ₹1,350** — SI = $(5000 \times 9 \times 3) \div 100 = ₹1,350$.
42. **B) 10%** — Loss = $3200 - 2880 = ₹320$; loss % = $320/3200 \times 100 = 10\%$.
43. **C) 220 km** — 2 h 45 min = 2.75 h; $80 \times 2.75 = 220$ km.
44. **C) 10 cups** — $\frac{5}{6} \times 12 = 10$ cups.
45. **C) 176** — $55\% \text{ of } 320 = 0.55 \times 320 = 176$.

Section 4 — Achievers Section (3 marks each)

46. **C) 300** — If the original number is x : $1.3x = 390 \rightarrow x = 300$.
47. **B) 33** — Three consecutive odd numbers $n-2$, n , $n+2$ sum to $3n = 99 \rightarrow n = 33$ (the middle number).
48. **C) ₹8,000** — SI = $(P \times 6 \times 5) \div 100 = 0.3 \times P = 2,400 \rightarrow P = 2,400 \div 0.3 = ₹8,000$.
49. **C) ₹3,000** — After rent, $\frac{3}{5}$ of the money remains. After groceries ($\frac{1}{3}$ of that spent), $\frac{2}{3}$ of $\frac{3}{5} = \frac{2}{5}$ of the original remains. So $\frac{2}{5}$ of the total = ₹1,200 $\rightarrow$ total = ₹3,000.
50. **C) 24** — Let the ages be $4x$ and $7x$. After 6 years: $(4x+6)/(7x+6) = \frac{5}{8} \rightarrow 8(4x+6) = 5(7x+6) \rightarrow 32x+48 = 35x+30 \rightarrow x = 6$. Younger's present age = $4 \times 6 = 24$.

Scoring guide (out of 60): 42+ is Level-2 cutoff range — strong. 30–41 is solid Level-1 performance. Below 30 — revisit the relevant topics before attempting Sample Paper 4.
